ARJUNA JEE 2027

Lecture -02 | Mathematics

TRIGONOMETRIC FUNCTIONS

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Topics *to be covered*

A Introduction to Trigonometry (Class-X)

B Angles & Quadrants

C Question Practice

D



Question Practice

Question

If $4x + 3y = 14$, then maximum value of x^4y^3 is 'a' which occurs at $x = b$ and $y = c$. Value of $(a + b + c)$ is (where $x, y \in R^+$)

A 126

B 128

C 130

D 132

$128 + 2 + 2$

$(0, \infty)$

x, x, x, x, y, y, y

$$\frac{4x + 3y}{7} \geq (x^4y^3)^{\frac{1}{7}}$$

$$\frac{14}{7} \geq (x^4y^3)^{\frac{1}{7}}$$

$$2^7 \geq (x^4y^3)$$

$$128 \geq x^4y^3$$

equality:

$x = y$

$4x + 3x = 14 \rightarrow x = 2 = y$

ACIDIC QUESTION

If $a_1, a_2, a_3 \ldots a_n \in R^+$ and $a_1 \cdot a_2 \cdot a_3 \ldots \ldots a_n = 1$ then show that minimum value of : $(1 + a_1 + a_1^2)(1 + a_2 + a_2^2) \ldots . (1 + a_n + a_n^2)$ is 3^n.

$R^+ \equiv (0, \infty)$

↳ +ve Real no.

$1, a_1, a_1^2 \rightarrow$ AM ≥ GM

$$\frac{1 + a_1 + a_1^2}{3} \geq (1 \times a_1 \times a_1^2)^{\frac{1}{3}}$$

$(a_1^3)^{\frac{1}{3}} = a_1$

$$1 + a_1 + a_1^2 \geq 3a_1$$
$$1 + a_2 + a_2^2 \geq 3a_2$$
$$\vdots$$
$$1 + a_n + a_n^2 \geq 3a_n$$

multiply

$$(1 + a_1 + a_1^2)(1 + a_2 + a_2^2) \ldots (1 + a_n + a_n^2)$$
$$\geq 3^n (a_1 a_2 a_3 \ldots a_n)$$
$$\geq 3^n$$

$a_1 a_2 a_3 \ldots a_n = 1$

HHPP.

$$\times \begin{cases} a \geq b \\ c \geq d \end{cases}$$
$$ac \geq bd$$

where

1. a, b, c, d → +ve

Question

$\sum_{r=1}^{99} r!\left(r^2+r+1\right)$ is equal to

A $102! - 100!$

B $100(100!) - 1$

C $99(100!) - 1$

D $100(99!) - 1$

$r! \times (r^2 + 2r + 1 - r)$

$r! \cdot ((r+1)^2 - r)$

$(r+1)^2 \cdot r! - r \cdot r!$

$T_r = (r+1).(r+1)! - r \cdot r!$

$(r+1)(r+1) \cdot r! =$

$(r+1) \cdot (r+1)!$

$T_1 = 2 \cdot 2! - 1 \cdot 1!$

$T_2 = 3 \cdot 3! - 2 \cdot 2!$

$T_3 = 4 \cdot 4! - 3 \cdot 3!$

$T_{99} = 100 \cdot 100! - 99 \cdot 99!$

$fnf = 100 \cdot 100! - 1$



EXPONENTIAL SERIES

RAR

$$\# \; e^x = \sum_{r=0}^{\infty} \frac{x^r}{r!} = \boxed{\frac{x^0}{0!} + \frac{x^1}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots \infty}$$

, $x \in R$

$$\# \; e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \ldots . \infty = \sum_{r=0}^{\infty} (-1)^r \frac{x^r}{r!}$$

used for alternate +/- sign.

$$\# \; \frac{e^x + e^{-x}}{2} = \left(\frac{1}{0!} + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \ldots \infty \right) \checkmark$$

$$\# \; \frac{e^x - e^{-x}}{2} = \left(\frac{x}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \ldots \infty \right) \checkmark$$

Acidic Question

30%

Find the sum of following series :

Remove n from N^r

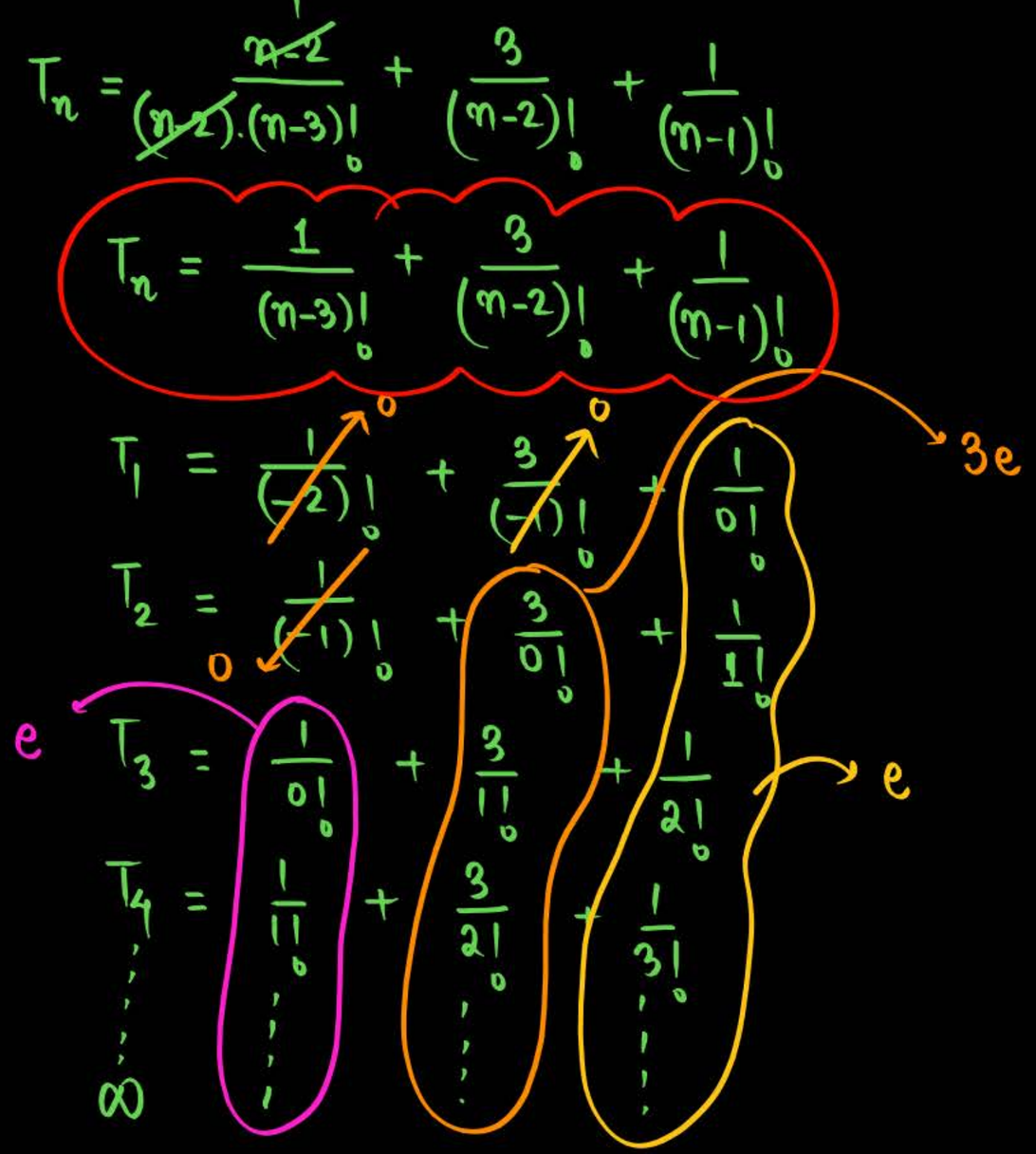
$$\sum_{n=1}^{\infty} \frac{n^3}{n!} = \frac{1^3}{1!} + \frac{2^3}{2!} + \frac{3^3}{3!} + \frac{4^3}{4!} + \ldots \infty = ke \qquad k = 5.$$

$$\frac{n^3}{n!} = \frac{n^2}{(n-1)!} = \frac{(n^2-1)+1}{(n-1)!} \quad \left((n-1)\cdot(n+1)\right)$$

$$= \frac{(n-1)\cdot(n+1)}{(n-1)(n-2)!} + \frac{1}{(n-1)!} = \frac{n-2+3}{(n-2)!} + \frac{1}{(n-1)!}$$

$$(n-2)\cdot(n-3)!$$

$$T_n = \frac{1}{(n-2)\cdot(n-3)!} + \frac{3}{(n-2)!} + \frac{1}{(n-1)!}$$



$$\# \ \frac{1}{(-1)!_0} = \frac{1}{(-2)!_0} = 0$$

$$fnf = e + 3e + e$$
$$= 5e.$$

Acidic Question

Find the sum of following series :

$$\frac{1}{0!}+\left(\frac{1+2}{1!}\right)+\left(\frac{1+2+3}{2!}\right)+\left(\frac{1+2+3+4}{3!}\right)+\ldots.\infty$$

$$T_n=\frac{1+2+3+\ldots+n}{(n-1)!}=\frac{n(n+1)}{2.(n-1)!}=\frac{1}{2}.\frac{n^2+n}{(n-1)!}$$

$(n-1)!=(n-1).(n-2)!$

$n=1,2,3,4\ldots.\infty$

$$T_n=\frac{1}{2}\left(\frac{n^2-1+1}{(n-1)!}+\frac{n-1+1}{(n-1)!}\right)$$

$$T_n = \frac{1}{2}\left(\frac{(n^2-1)+1}{(n-1)!_0} + \frac{(n-1)+1}{(n-1)!_0}\right)$$

$(n-1)(n+1)$

$$\frac{n^2-1^2}{(n-1)!_0} + \frac{1}{(n-1)!_0} + \frac{n-1}{(n-1)!_0} + \frac{1}{(n-1)!_0}$$

$(n-1)(n-2)!_0$

$(n-1).(n-2)!_0$

$$\frac{n+1}{(n-2)!_0} + \frac{1}{(n-1)!_0} + \frac{1}{(n-2)!_0} + \frac{1}{(n-1)!_0}$$

$$T_n = \frac{1}{(n-3)!_0} + \frac{3}{(n-2)!_0} + \frac{1}{(n-1)!_0} + \frac{1}{(n-2)!_0} + \frac{1}{(n-1)!_0}$$

$$\frac{n+1}{(n-2)!_0} = \frac{n-2}{(n-2)!_0} + \frac{3}{(n-2)!_0}$$

$(n-2)(n-3)!_0$

$$= \frac{1}{(n-3)!_0} + \frac{3}{(n-2)!_0}$$

$$T_n = \frac{1}{2}\left(\frac{1}{(n-3)!} + \frac{3}{(n-2)!} + \frac{1}{(n-1)!} + \frac{1}{(n-2)!} + \frac{1}{(n-1)!}\right)$$

$$T_n = \frac{1}{2}\left(\frac{1}{(n-3)!} + \frac{4}{(n-2)!} + \frac{2}{(n-1)!}\right)$$

$n = 1, 2, 3, \ldots \infty$

$n = 3, 4, 5 \ldots \infty$ $\quad$ $n = 2, 3, 4 \ldots \infty$

$$\left(\frac{1}{0!} + \frac{1}{1!} + \frac{1}{2!} + \ldots \infty\right)$$

$$f_{nf} = \frac{1}{2}(e + 4e + 2e) = \frac{7e}{2}$$

Question [JEE Mains-2023]

The sum $\sum_{n=1}^{\infty} \frac{2n^2+3n+4}{(2n)!}$ is equal to



A $\frac{11e}{2}+\frac{7}{2e}$

B $\frac{13e}{4}+\frac{5}{4e}-4$

C $\frac{11e}{2}+\frac{7}{2e}-4$

D $\frac{13e}{4}+\frac{5}{4e}$



LOGARITHMIC SERIES

$$\ln(1-x) = -\frac{x^1}{1} - \frac{x^2}{2} - \frac{x^3}{3} - \frac{x^4}{4} - \frac{x^5}{5} \ldots\ldots\infty, \qquad -1 \le x < 1$$

$$\ln(1+x) = \frac{x^1}{1} - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} \ldots\ldots\infty, \qquad -1 < x \le 1 \checkmark$$

Question

$$1+\frac{2}{3}-\frac{2}{4}+\frac{2}{5}-\ldots\ldots\infty=$$

A $\log_e 3$

B $\log_e 4$

C $\log_e\left(\frac{e}{2}\right)$

D $\log_e\left(\frac{2}{3}\right)$

$$1+2\left(\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\frac{1}{6}+\ldots\infty\right)$$

$$1+2\left(\ln 2-\frac{1}{2}\right)$$

$$\cancel{1}+2\ln 2-\cancel{1}=\log_e 2^2=\log_e 4=\ln 4.$$

$$\ln(1+x)=\frac{x^1}{1}-\frac{x^2}{2}+\frac{x^3}{3}-\frac{x^4}{4}+\frac{x^5}{5}\ldots$$

$x=1$

$$\ln 2=\left(\frac{1}{1}-\frac{1}{2}\right)+\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\frac{1}{6}+\ldots$$

Question

Evaluate the sum of the series $\frac{1}{1\cdot2} - \frac{1}{2\cdot3} + \frac{1}{3\cdot4} - \frac{1}{4\cdot5} + \ldots\ldots\infty$

$\frac{1}{1\times2} = 1 - \frac{1}{2}$

$\frac{1}{2\times3} = \frac{1}{2} - \frac{1}{3}$

$\frac{1}{3\times4} = \frac{1}{3} - \frac{1}{4}$

$\left(1 - \frac{1}{2}\right) - \left(\frac{1}{2} - \frac{1}{3}\right) + \left(\frac{1}{3} - \frac{1}{4}\right) - \left(\frac{1}{4} - \frac{1}{5}\right) + \ldots\infty$

$1 - \frac{1}{2} - \frac{1}{2} + \frac{1}{3} + \frac{1}{3} - \frac{1}{4} - \frac{1}{4} + \frac{1}{5} + \ldots\infty$

$1 - \frac{2}{2} + \frac{2}{3} - \frac{2}{4} + \frac{2}{5} - \frac{2}{6} + \ldots\infty$

$2\left(\frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \frac{1}{6} + \ldots\infty\right) = 2\left(\ln 2 - \frac{1}{2}\right)$

$= 2\ln 2 - 1.$

$$\text{Ans} = \underbrace{2\ln 2}_{1} - \frac{1}{1} \cdot \checkmark$$

$$= \ln 4 - \log_e e \checkmark$$

$$= \ln\left(\frac{4}{e}\right) \checkmark$$

Question

$$\frac{1}{5}+\frac{1}{2}\cdot\frac{1}{5^2}+\frac{1}{3}\cdot\frac{1}{5^3}+\ldots\ldots\infty=$$

A $\log_e \frac{4}{5}$

B $\log_e \frac{\sqrt{5}}{2}$

C $2\log_e \frac{\sqrt{5}}{2}$

D None of these

$$-\ln(1-x)=\frac{x^1}{1}+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+\ldots\infty$$

$x=\frac{1}{5}$.

$$-\ln\left(1-\frac{1}{5}\right)=\frac{1}{1}\times\frac{1}{5}+\frac{1}{2}\times\left(\frac{1}{5}\right)^2+\frac{1}{3}\times\left(\frac{1}{5}\right)^3+\ldots\infty.$$

$2\log_e \frac{\sqrt{5}}{2}=\log_e\left(\frac{\sqrt{5}}{2}\right)^2$

$\ln\left(\frac{4}{5}\right)^{-1}=-\ln\left(\frac{4}{5}\right)=$

$\ln\left(\frac{5}{4}\right)$.

Challenger Question [JEE Mains-2021]

If $0 < x < 1$, then $\frac{3}{2}x^2 + \frac{5}{3}x^3 + \frac{7}{4}x^4$ ....., is equal to



A $x\left(\frac{1+x}{1-x}\right) + \log_e(1-x)$

B $x\left(\frac{1-x}{1+x}\right) + \log_e(1-x)$

C $\frac{1-x}{1+x} + \log_e(1-x)$

D $\frac{1+x}{1-x} + \log_e(1-x)$



SEQUENCE & SERIES

BLOCK-I : Questions on T_n & S_n of A.P. & G.P.
T_n of H.P. and Inserting means between two numbers.

BLOCK-II : A.G.P. (Single Layer & Multiple Layer)

BLOCK-III : **MISCELLANEOUR SERIES**
Questions on $\sum n$, $\sum n^2$ & $\sum n^3$.
Write T_n & Apply Sigma.
T_n by observation or method of subtraction (Tricks)



SEQUENCE & SERIES

BLOCK-IV : T.M.O.D.

Type-01 : Terms of A.P. are in multiplication in Numerator or Denominator.

Type-02 : Bheja Fry (Based on Factorisation)

BLOCK-V : Inequality (AM $\geq$ GM $\geq$ HM) & Cauchy Schwarz

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BLOCK-VI : Factorial Series (T.M.O.D.)

Exponential Series

log-series

TRIGONOMETRIC FUNCTIONS (RATIOS & IDENTITIES)



BASIC OF TRIGONOMETRY (CLASS-X)

Six Trigonometric Functions:

$$\sin(\theta) = \frac{P}{H} \qquad \operatorname{cosec}(\theta) = \frac{H}{P}$$

$$\cos(\theta) = \frac{B}{H} \qquad \sec(\theta) = \frac{H}{B}$$

$$\tan(\theta) = \frac{P}{B} \qquad \cot(\theta) = \frac{B}{P}$$

Trigonometric Identities:

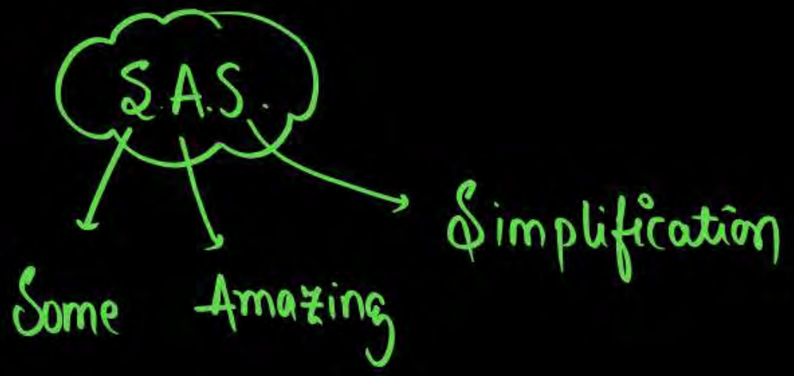
(i) $\sin^2\theta + \cos^2\theta = 1$

(ii) $\sec^2\theta - \tan^2\theta = 1$ or

$(\sec\theta - \tan\theta)(\sec\theta + \tan\theta) = 1$

(iii) $\operatorname{cosec}^2\theta - \cot^2\theta = 1$ or

$(\operatorname{cosec}\theta - \cot\theta)(\operatorname{cosec}\theta + \cot\theta) = 1$



S.A.S → Ques^{n}. → sin & cos

RAPID FIRE DIBY'S (CLASS-X)

$\sqrt{\square^2} = |\square|$

1.

Which of the following reduces to unity for $0° < A < 90°$

(1) $(\sec^2 A - 1)\cot^2 A$

(2) $\cos A \operatorname{cosec} A \sqrt{\sec^2 A - 1}$

(3) $(\operatorname{cosec}^2 A - 1)\tan^2 A$

(4) $(1 - \cos^2 A)(1 + \cot^2 A)$

(5) $\sin A \sec A \sqrt{\operatorname{cosec}^2 A - 1}$

(6) $(1 + \tan^2 A)(1 - \sin^2 A)$

(7) $\sec^2 A - \sin^2 A \sec^2 A$

(8) $\frac{1}{1+\sin^2 A} + \frac{1}{1+\operatorname{cosec}^2 A}$

(9) $\frac{\tan^2 A \sin^2 A}{\tan^2 A - \sin^2 A}$

(10) $\frac{\cot^2 A \cos^2 A}{\cot^2 A - \cos^2 A}$

$\sqrt{\tan^2 A} = |\tan A|$ (+ve)

(2) $\cos A \cdot \frac{1}{\sin A} \sqrt{\tan^2 A} = \cos A \cdot \frac{1}{\sin A} \cdot \frac{\sin A}{\cos A} = 1$

(8) $\frac{1}{1+\sin^2 A} + \frac{1}{1+\operatorname{cosec}^2 A}$ → $\frac{1}{\sin^2 A}$

$$\frac{1}{1+\sin^2 A} + \frac{1}{1+\frac{1}{\sin^2 A}}$$

$$\frac{1}{1+\sin^2 A} + \frac{\sin^2 A}{\sin^2 A + 1} = \frac{1+\sin^2 A}{1+\sin^2 A} = 1 \checkmark$$

(10) $\dfrac{\cot^2 A \cdot \cos^2 A}{\cot^2 A - \cos^2 A} = \dfrac{\cos^2 A \cos^2 A}{\sin^2 A \left(\dfrac{\cos^2 A}{\sin^2 A} - \cos^2 A\right)} = \dfrac{\cos^2 A \cos^2 A}{\sin^2 A \cos^2 A \left(\dfrac{1}{\sin^2 A} - 1\right)}$

$\dfrac{1}{\sin^2 A} - 1 = \dfrac{1 - \sin^2 A}{\sin^2 A}$

$= \dfrac{\cos^2 A}{\sin^2 A}$

$= \dfrac{\cos^2 A}{\sin^2 A \cdot \dfrac{\cos^2 A}{\sin^2 A}} = 1$

Question

(i) Prove that : $\sin^4\theta + \cos^4\theta = 1 - 2\sin^2\theta\cos^2\theta$

(ii) Prove that : $\sin^6\theta + \cos^6\theta = 1 - 3\sin^2\theta\cos^2\theta$

RAR

$\sin^2\theta + \cos^2\theta = 1$.

SBS.

$\sin^4\theta + \cos^4\theta + 2\sin^2\theta\cos^2\theta = 1$.

Cube both side.

$(\sin^2\theta + \cos^2\theta)^3 = 1^3$

$(\sin^2\theta)^3 + (\cos^2\theta)^3 + 3\sin^2\theta\cos^2\theta\underbrace{(\sin^2\theta + \cos^2\theta)}_{1} = 1$.

$\sin^6\theta + \cos^6\theta = 1 - 3\sin^2\theta\cos^2\theta$.

Question [JEE Mains-2019]

73%

Let $f_k(x) = \frac{1}{k}\left(\sin^k x + \cos^k x\right)$ where $x \in R$ and $k \geq 1$. Then $f_4(x) \ominus f_6(x)$ equals:

A $\frac{1}{4}$

B $\frac{1}{12}$

C $\frac{1}{6}$

D $\frac{1}{3}$

$k=4$

$f_4(x) = \frac{1}{4}\left(\sin^4 x + \cos^4 x\right)$

$= \frac{1}{4}\left(1 - 2\sin^2 x \cdot \cos^2 x\right) = \boxed{\frac{1}{4} - \frac{1}{2}\sin^2 x \cdot \cos^2 x}$

$k=6$

$f_6(x) = \frac{1}{6}\left(\sin^6 x + \cos^6 x\right) = \frac{1}{6}\left(1 - 3\sin^2 x \cdot \cos^2 x\right)$

$= \boxed{\frac{1}{6} - \frac{1}{2}\sin^2 x \cdot \cos^2 x}$

$f_4(x) \ominus f_6(x)$

$\left(\frac{1}{4} - \frac{1}{2}\sin^2 x \cos^2 x\right) - \left(\frac{1}{6} + \frac{1}{2}\sin^2 x \cos^2 x\right)$

$\frac{1}{4} - \frac{1}{6} = \frac{2}{24}$

$= \frac{1}{12}$ ✓

Question [JEE Mains-2013]

The expression $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}$ can be written as:



A sin A cos A + 1

B sec A cosec A + 1

C tan A + cot A

D sec A + cosec A

Acidic Question

The value of $\log_{\sin^2 x+\cos^4 x+2}(\cos^2 x+\sin^4 x+2)$ is equal to

A 1

B −1

C 0

D 2

$A = \cos^2 x + \sin^4 x + 2 = \cos^2 x + (1-\cos^2 x)^2 + 2$

$= \cos^2 x + 1 + \cos^4 x - 2\cos^2 x + 2$

$= \cos^4 x + 1 - \cos^2 x + 2$

$= \cos^4 x + \sin^2 x + 2$

$= B$

$\# \log_A A = 1$

$\sin^4 x = (\sin^2 x)^2 = (1-\cos^2 x)^2$

Question

67%

Suppose that $a \cos\theta = b$ and $c \sin\theta = d$, for θ and some constants a, b, c, d. then which of the following is true?

A $a^2c^2 = b^2c^2 + a^2d^2$

B $a^2d^2 = b^2c^2 + a^2c^2$

C $b^2c^2 = a^2d^2 + a^2c^2$

D $a^2b^2 = b^2c^2 + c^2d^2$

$\cos\theta = \frac{b}{a}$ $\sin\theta = \frac{d}{c}$

$\sin^2\theta + \cos^2\theta = 1$

$\left(\frac{d}{c}\right)^2 + \left(\frac{b}{a}\right)^2 = 1.$

$a^2d^2 + c^2b^2 = a^2c^2$

$\frac{d^2}{c^2} + \frac{b^2}{a^2} = 1$

Challenger Question

Questions are based on elimination of θ :

(i) If $\operatorname{cosec}\theta - \sin\theta = m$ and $\sec\theta - \cos\theta = n$, then $(m^2n)^{2/3} + (mn^2)^{2/3} = 1$.

Smartest Method $\theta = 45^\circ \Rightarrow m = \sqrt{2} - \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{2}} = n$

$\frac{1}{\sin\theta} - \sin\theta = m$

$\frac{1-\sin^2\theta}{\sin\theta} = m$

$\boxed{\frac{\cos^2\theta}{\sin\theta} = m}$ —①

$\frac{1}{\cos\theta} - \cos\theta = n$

$\frac{1-\cos^2\theta}{\cos\theta} = n$

$\boxed{\frac{\sin^2\theta}{\cos\theta} = n}$ —②

$(m^3)^{\frac{2}{3}} + (n^3)^{\frac{2}{3}}$

$m^2 + n^2$

$\frac{1}{2} + \frac{1}{2}$

$$\frac{\cos^2\theta}{\sin\theta} = m \quad -(1)$$

$$\frac{\sin^2\theta}{\cos\theta} = n \quad -(2)$$

$$\frac{\cos^2\theta}{m} = \sin\theta$$

$$\left(\left(m^2 n\right)^{\frac{1}{3}}\right)^{\frac{2}{3}} = \sin\theta$$

$$m$$

$$\frac{m^{\frac{4}{3}} n^{\frac{2}{3}}}{m'} = \sin\theta = \left(m^{\frac{1}{3}} n^{\frac{2}{3}}\right)$$

$$\sin\theta = \left(m n^2\right)^{\frac{1}{3}}$$

$$\sin^2\theta = n\cos\theta$$

$$\left(\frac{\cos^2\theta}{m}\right)^2 = n\cos\theta$$

$$\frac{\cos^4\theta}{m^2} = n\cos\theta$$

$$\cos^3\theta = m^2 n$$

$$\cos\theta = \left(m^2 n\right)^{\frac{1}{3}}$$

$$\cos^2\theta + \sin^2\theta = 1$$

$$\left(\left(m^2 n\right)^{\frac{1}{3}}\right)^2 + \left(\left(m n^2\right)^{\frac{1}{3}}\right)^2 = 1$$

$$\left(m^2 n\right)^{\frac{2}{3}} + \left(m n^2\right)^{\frac{2}{3}} = 1.$$

Challenger Question

Questions are based on elimination of θ :

(i) If $\operatorname{cosec}\theta - \sin\theta = m$ and $\sec\theta - \cos\theta = n$, then $(m^2n)^{2/3} + (mn^2)^{2/3} = 1$

$\frac{1}{\sin\theta} - \sin\theta = m$

$\frac{1-\sin^2\theta}{\sin\theta} = m$

$\frac{\cos^2\theta}{\sin\theta} = m$ —①

$\frac{1}{\cos\theta} - \cos\theta = n$

$\frac{1-\cos^2\theta}{\cos\theta} = n$

$\frac{\sin^2\theta}{\cos\theta} = n$ —②

$\frac{\cos^4\theta}{\sin^2\theta} \times \frac{\sin^2\theta}{\cos\theta}$

$(\cos^3\theta)^{\frac{2}{3}}$

$\cos^2\theta + \sin^2\theta$

Challenger Question

Questions are based on elimination of θ :

(i) If $\operatorname{cosec}\theta - \sin\theta = m$ and $\sec\theta - \cos\theta = n$, then $(m^2 n)^{2/3} + (mn^2)^{2/3} =$

(ii) If $a\sec\theta - c\tan\theta = d$ & $b\sec\theta + d\tan\theta = c$, then show $a^2 + b^2 = c^2 + d^2$



Question

If $a\cos\theta + b\sin\theta = 3$ and $a\sin\theta - b\cos\theta = 4$, then $a^2 + b^2$ has the value

A 25

B 14

C 7

D 15

SBS

$a^2\cos^2\theta + b^2\sin^2\theta + 2ab\sin\theta\cos\theta = 9$

$a^2\sin^2\theta + b^2\cos^2\theta - 2ab\sin\theta\cos\theta = 16$

add

$a^2(\cos^2\theta+\sin^2\theta) + b^2(\sin^2\theta+\cos^2\theta) = 25$

$a^2+b^2 = 25$

youtube

$\theta = 90^\circ$

$\cos 90^\circ = 0$

$\sin 90^\circ = 1$

$b = 3$ & $a = 4$



BASIC OF TRIGONOMETRY (CLASS-X)

Point-03: Trigonometric Values

	0°	30°	45°	60°	90°
sin θ					
cos θ					
tan θ					
cot θ					
sec θ					
cosec θ					

DO IT BY YOURSELF (DIBY)

INEQUALITY

DIBY-103

Minimum value of sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ & a^{10} (where $a > 0$)

[IIT-JEE-2011]

[Ans. 8]

DIBY-104

If $a > 0, b > 0, c > 0$, then show that : $(a+b+c)\left(\frac{1}{a}+\frac{1}{b}+\frac{1}{c}\right) \geq 9$ [JEE-1984]

DIBY-105

For positive real number a, b & c show that $\left((1+a)(1+b)(1+c)\right)^7 > 7^7 a^4 b^4 c^4$ [JEE-2004]

DIBY-106

Let $x, y > 0$. If $x^3y^2 = 2^{15}$, then the least value of $3x + 2y$ is [JEE Mains-2022]

(A) 30 (B) 32 (C) 36 (D) 40

[Ans. (D)]

INEQUALITY

DIBY-107

The minimum value of $f(x) = a^{a^x} + a^{1-a^x}$,, where $a, x \in R$ and $a > 0$, is equal to

[JEE Mains-2021]

(A) $a + \frac{1}{a}$ (B) $2a$ (C) $a + 1$ (D) $2\sqrt{a}$

[Ans. D]

DIBY-108

Let a, b, c and d be positive real numbers such that $a + b + c + d = 11$. If the maximum value of $a^5b^3c^2d$ is 3750β, then the value of β is [JEE Mains-2023]

(A) 110 (B) 108 (C) 90 (D) 55

[Ans. C]

DIBY-109

If the minimum value of $f(x) = \frac{5x^2}{2} + \frac{\alpha}{x^5}, x > 0$ is 14, then the value of α is equal to

[JEE Mains-2022]

(A) 32 (B) 64 (C) 128 (D) 256

[Ans. (C)]

EXPONENTIAL & LOG SERIES

DIBY-110

Simplify : $S = \frac{2}{1!} + \frac{4}{3!} + \frac{6}{5!} + \frac{8}{7!} + \ldots\ldots \infty$

[Ans. (e)]

DIBY-111

$$S = 1 + \frac{1+2}{2!} + \frac{1+2+2^2}{3!} + \frac{1+2+2^2+2^3}{4!} + \ldots\ldots \infty$$

[Ans. $(e^2 - e)$]

DIBY-112

The sum $1 + \frac{1+3}{2!} + \frac{1+3+5}{3!} + \frac{1+3+5+7}{4!} + \ldots\ldots$ upto ∞ terms, is equal to

[JEE Mains-2025]

(A) 6e (B) 4e (C) 3e (D) 2e

[Ans. (D)]

EXPONENTIAL & LOG SERIES

DIBY-113

The sum of the series $\sum_{n=1}^{\infty} \frac{n^2+6n+10}{(2n+1)!}$ is equal to: [JEE Mains-2021]

(A) $\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$

(B) $\frac{41}{8}e - \frac{19}{8}e^{-1} - 10$

(C) $\frac{41}{8}e + \frac{19}{8}e^{-1} + 10$

(D) $-\frac{41}{8}e + \frac{19}{8}e^{-1} - 10$

[Ans. : B]

DIBY-114

If $0 < x < 1$ and $y = \frac{1}{2}x^2 + \frac{2}{3}x^3 + \frac{3}{4}x^4 + \cdots$, then the value of e^{1+y} at $x = \frac{1}{2}$ is: [JEE Mains-2021]

(A) $2e^2$

(B) $\frac{1}{2}e^2$

(C) $2e$

(D) $\frac{1}{2}\sqrt{e}$

[Ans. : B]



Homework

Re-attempt all Class Examples

DPP & DIBY

Module: Only 2 Exercise

Topicwise & Bridge(Setu)- 4 sittings each of 1 HR

It is never about end result

It's always about Journey

Thank You

#futureIITians